

HOLC Redlining Maps and Present-Day Geography

A form filled in on the edge of Cleveland in 1939, and the ground it still describes

Democracy's Data

Last updated 31 August 2026

An appraiser working for the federal government walked a few streets on the southern edge of Cleveland in the late summer of 1939 and then filled in a form. The form had spaces for the terrain, the class of the inhabitants, the age of the buildings, the availability of mortgage money. Under the heading *Clarifying Remarks* he wrote:

This area is the least desirable neighborhood in Garfield Heights. Immediate proximity to negro settlement abutting north boundary – absence of transportation – lack of sewers – unpaved streets and general isolation all have taken their toll. This community was developed about 20-25 years ago and then stopped with the influx of negroes into abutting area. At present, property can be sold only at sacrifice and mortgage money is not available which indicates that potentially this neighborhood will eventually be taken over by negroes with improvement in their financial condition. Approximately 30 homes are located in this area; developed about 40%; no further development is anticipated for white ownership.

The area is D9, *Garfield Hts. (Orchard Ave. Section)*, and the sheet is dated 9-9-39. Two other lines on that form matter as much as the paragraph. One is headed *Infiltration of*, and on it he wrote *Potential influx of Negro*. The other asks what share of the residents were Black, and on that line he wrote 0, meaning none. He gave the area a grade of D.

Race was not standing in for something else on that sheet. **It was the stated criterion**, applied to a street where by the appraiser's own count no Black family yet lived. Grades like this one told a federal corporation where mortgage lending was safe, and mortgage lending is how most American families come to own anything at all. School money is raised on property. Services, transit and policing are handed out on a map. So the question is worth asking plainly: ninety years later, on the actual ground, does that line still show?

01 The test

Answering it means putting two files beside each other. The first is *Mapping Inequality*, the digitized residential security maps of the Home Owners' Loan Corporation. It holds 10,154 neighborhood areas traced from 314 city surveys of 1935 to 1940, 9,324 of them carrying a grade, and it is published by the Digital Scholarship Lab at the University of Richmond (<https://dsl.richmond.edu/panorama/redlining/>). The second is the 2020 Census, which counts every resident of every census tract (https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/).

The 1930s file is the right instrument because it is not a study of the period. It is the working paperwork of the agency that did the sorting, written to be acted on rather than read later. The men who wrote it had no reason to overstate the part race played, because they did not think it needed justifying. A modern index of which neighborhoods were disfavored would carry its author's theory of what disfavor looks like. Nobody graded on these sheets was asked anything, and most never learned that a federal agency had given their street a letter. That is what makes it the kind of record Section III is about.

The census is doing a second job here, and it is easy to miss. The grade file holds shapes and letters and no people at all. Every share below is a count of Black residents over a count of all residents, and both of those numbers are borrowed from the 2020 census. The administrative record supplies only the letter. Somebody else counted everyone standing under it.

02 The data

The table the figures rest on is `tracts.csv`: one row per 2020 census tract, 4,489 rows, covering 56 cities in 6 states. Each row carries `GE0ID`, the tract's eleven-digit identifier, with `state` and `city`; `grade`, the letter the ground under it was given in the 1930s; and `total`, `white`, `black` and `pct_black` from the census.

Tract GE0ID	State	City	1930s HOLC grade	2020 population	Black residents	% Black
06001400100	CA	Oakland	B	3038	144	4.74
06001400200	CA	Oakland	B	2001	40	2.00
06001400300	CA	Oakland	C	5504	561	10.19

A single row of that file is a strange object, and the letter in it is the strangest part. The corporation used four:

Grade	Label	Color
A	Best	green
B	Still Desirable	blue
C	Definitely Declining	yellow
D	Hazardous	red

The D areas were printed in red, which is where the word *redlining* comes from, and the paragraph at the top of this brief is what a D looked like from the inside. Figure 1 is Cleveland's whole sheet.

HOLC grade, 1939

- A Best (31 areas, 11% of graded land)
- B Still Desirable (55 areas, 23% of graded land)
- C Definitely Declining (76 areas, 47% of graded land)
- D Hazardous (28 areas, 18% of graded land)

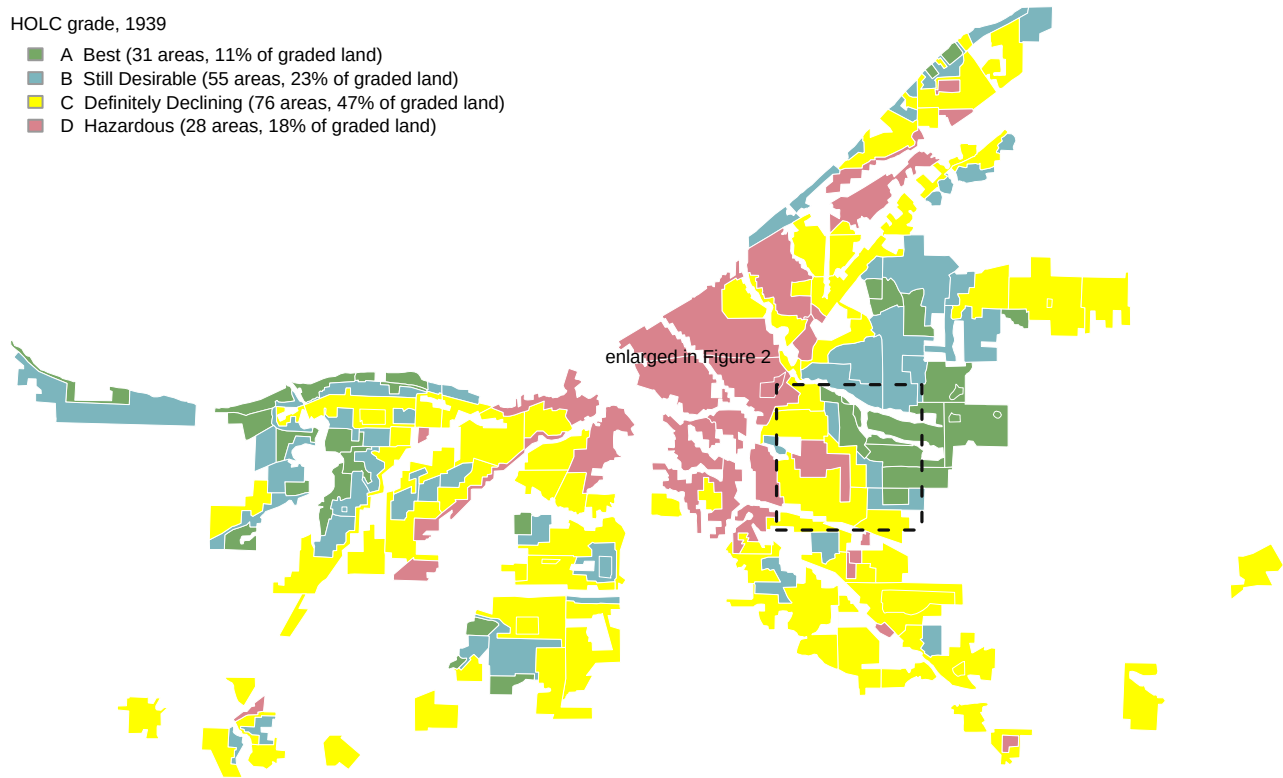


Figure 1. Cleveland as the Home Owners' Loan Corporation graded it in 1939. 190 graded areas over 302 square kilometers, filled with the corporation's own four colors read out of the source file rather than chosen here. A map is the right form because the grade was applied to ground. Red sits in the industrial core, an average of 7.8 km from the center of the surveyed area; green sits 12.4 km out at the edge. The concentric rings of creditworthiness are visible before any census number arrives. The dashed rectangle marks the ground enlarged in Figure 2.

Getting the census onto that map takes a decision, because the two sets of shapes have nothing to do with each other. A 1937 appraiser's boundary was drawn with no knowledge of a 2020 census tract, and tracts have been redrawn every decade since. There is no shared identifier and no lookup table. The rule used here is the simplest one that can be defended: take each tract's center point, and ask which graded area contains it.

Step	Value
HOLC areas digitized	10,154
of them graded A to D	9,324
Rule applied	tract center point inside a graded polygon
Tracts matched to a grade	4,489
Cities represented	56
States	6

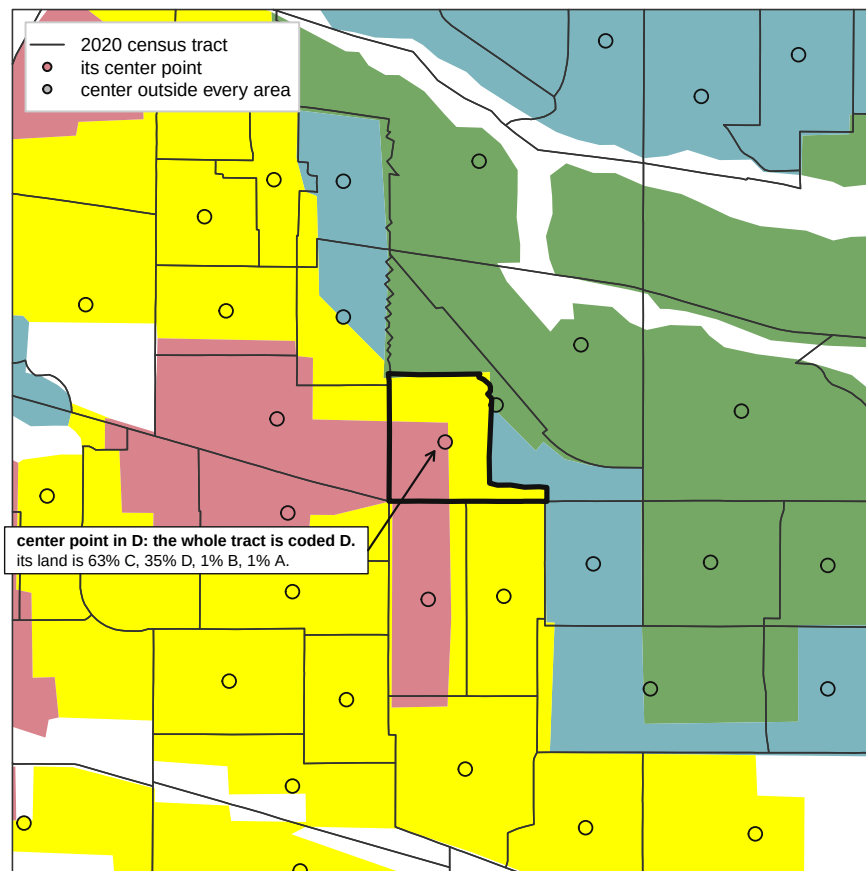


Figure 2. A 5.6 km window of Cleveland, with 1939 underneath and 2020 on top. Colored surfaces are HOLC areas in the corporation's own grade colors, thin outlines are 2020 census tracts, and each dot is the tract center point that the rule consults. The heavily outlined tract is the rule getting it wrong: its center falls in a D area, so the whole tract is coded D, but its land is 63% C, 35% D, 1% B, 1% A.

That is the cost, and it is worth stating in numbers rather than leaving to one picture. Among the 587 tracts the rule grades in Cleveland and Philadelphia, 33 get a letter that is not the letter covering most of their own land. A tract whose center lands outside every graded area is dropped however much of it overlaps one, and 181 tracts are dropped that way. What the rule is, at least, is reproducible: re-running it from the polygons returns the grade the file already carries for all but 1 of the 588 tracts in those two cities. Choosing between rules of this kind is a subject of its own, and Join Keys and Crosswalks, in Section IV, is about it. A crosswalk is a table that translates one set of areas into another, and one weighted by area or by population would grade some of these tracts differently.

Three cleaning decisions sit under all of it. The grade lives in a field called `grade`, and the more natural guess, `holic_grade`, exists in other distributions of these maps and returns nothing here, which would produce a complete, plausible, entirely ungraded map. GEOID has to be read as text, because the Californian identifiers begin with a zero that a numeric read deletes. And the source's own area field is in square degrees, which is not a unit of distance anywhere, so every area and overlap here is computed after reprojecting to metres.

Before you read on, commit to two numbers. The file holds 33 cities with both A-graded and D-graded ground. **In the average city, how many percentage points more Black is the D-graded ground than the A-graded ground today?** And **how many of the 33 cities run the other way?** Write both down.

03 What it says about democracy

Pool all 4,489 tracts by grade and the obvious comparison appears at once.

HOLC grade	Tracts	2020 population	Black residents	% Black
A	273	978,270	163,372	16.7
B	899	3,269,467	762,053	23.3
C	2096	7,273,922	1,555,044	21.4
D	1221	3,984,323	1,018,080	25.6

Areas graded A are 16.7% Black today and areas graded D are 25.6%, a gap of 8.9 points. After the appraisal sheet at the top of this brief that is a deflating number, and the ordering does not even behave: C areas are 21.4% Black, less than the 23.3% in B areas.

The pooled table is not measuring grades. It is measuring cities. Cleveland's D areas and Oakland's A areas sit in it together, and American cities differ enormously in overall Black population for reasons that have nothing to do with the corporation. San Francisco's D neighborhoods sit in a city that is about 5% Black; Cleveland's A neighborhoods sit in a city that is roughly half Black. Differences between cities swamp the difference within them, and the difference within them is the only one the question is about.

So make the comparison inside each city, where the same housing market, the same city government, the same school district boundaries and the same regional history apply to both grades.

City	A tracts	D tracts	A areas % Black	D areas % Black	Gap
Cleveland	25	47	21.3	68.1	46.9
Pontiac	1	4	32.1	71.0	39.0
Decatur	1	4	19.4	54.8	35.4
Toledo	3	4	6.6	40.5	33.9
Greater Kansas City	5	34	4.0	37.7	33.6
St. Louis	15	17	14.0	47.6	33.6
Chicago	18	249	3.2	36.6	33.4
Pittsburgh	7	43	4.2	34.1	29.9

In Cleveland, the city whose 1939 sheet opens this brief, neighborhoods graded A are 21.3% Black today and neighborhoods graded D are 68.1%. That is 46.9 points between

two sets of neighborhoods separated by nothing except a letter assigned before the Second World War. Cleveland is the extreme case, and the extreme case is never the argument. Across all 33 cities holding both grades the average gap is 14.5 points, and 26 of the 33 run in the direction the history predicts.

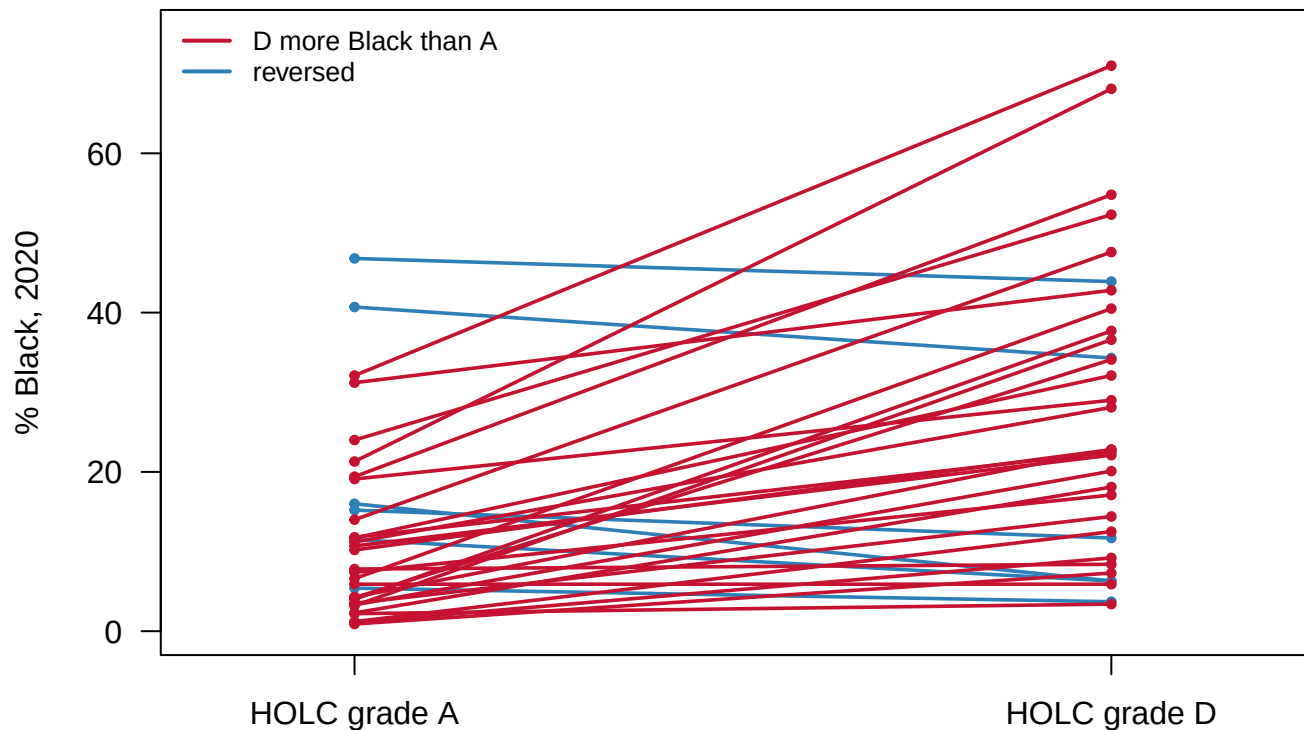


Figure 3. Within each city, from its A-graded ground to its D-graded ground. One line per city, joining the share of residents who are Black in the tracts graded A to the share in the tracts graded D, 2020 Census. Of the 33 cities with both grades, 26 rise and 6 fall; the steepest rise is Cleveland, at 46.9 points. The average line rises 14.5 points.

The 6 downward lines are the most useful thing in the figure, and they are two different things wearing one sign.

City	A tracts	D tracts	A population	D population	A % Black	D % Black	Gap
Joliet	2	1	6,265	3,040	16.0	6.3	-9.6
Philadelphia	28	138	97,336	548,467	40.7	34.3	-6.4
Stockton	1	6	5,750	26,692	11.5	6.3	-5.2
Canton	1	1	3,278	2,884	15.2	11.7	-3.5
Detroit	22	102	82,787	251,281	46.8	43.9	-3.0
San Jose	1	6	3,728	29,341	5.4	3.7	-1.7

Philadelphia is a real reversal. Its D areas are 34.3% Black and its A areas 40.7%, backwards by 6.4 points across 138 D tracts holding 548,467 people. North Philadelphia was graded D and has been substantially gentrified, and several historically Black neighborhoods there were never graded D at all. A great deal happened between 1937 and 2020: urban renewal, deindustrialization, highway construction, and lately some very uneven reinvestment.

Joliet is not a real reversal. Its 9.6-point gap rests on 2 A tracts against 1 D tract, and a percentage computed on a handful of tracts is not a stable quantity. Printing it beside Cleveland's 25-against-47 as though the two carry equal weight would mislead a reader with entirely accurate numbers. Several of the other reversals are small in the same way.

So the sentence this evidence will bear is the middle one of three. The grade is associated with a neighborhood's racial composition today, on average, within a city. It did not determine it, because Philadelphia went the other way. And it has not been shown here to have caused it.

04 What this brief cannot tell you

It cannot tell you whether the maps did anything. The appraisers were describing segregation that already existed, much of it produced by restrictive covenants, by violence and by private lending that predated the corporation. On that reading these maps are an excellent record of 1930s segregation and no evidence at all about its causes. Separating the two needs a different design, not a bigger version of this one. Aaronson, Hartley and Mazumder (2021) compared similar neighborhoods in cities that just barely were mapped and just barely were not. They found effects that lasted decades.

It cannot speak for the country. The corporation surveyed cities above a population threshold, so there was never a national frame, and this file holds 56 cities in 6 states.

It cannot show a path. This is one snapshot. These neighborhoods may have been converging for fifty years, or diverging, or crossing over, and a single 2020 cross-section looks the same in all three cases.

And a grade is not a measurement. It is one appraiser's letter, with no second rater, no units and no standard error, and it is being used here as though it were an assignment somebody made at random. Grade was never the only thing that differed between an A neighborhood and a D neighborhood in 1937.

05 What you should have learned

- Within a city, neighborhoods the federal government graded "Hazardous" in the 1930s are on average 14.5 points more Black today than neighborhoods the same government graded "Best", across 33 cities and 4,489 tracts.
- The pooled national comparison is the wrong test and it understates: it puts A areas in one city against D areas in another, and it returns a gap of only 8.9 points with the middle grades out of order.
- 6 of the 33 cities run the other way, and they are not all the same thing: Philadelphia reverses across 138 D tracts, while Joliet reverses on 1.
- The link between 1939 and 2020 is one sentence of rule, and it has a measurable cost: 33 of 587 graded tracts in Cleveland and Philadelphia carry a letter that is not the letter covering most of their land.
- The record supplies the grade and nothing else. Every person in every share above was counted by the census, not by the corporation.

06 Extensions

None of these is answered above, and every one can be answered from the tables the Sources section hands over.

- `cities.csv` gives each city a gap, the share Black in A areas against the share in D areas. Sort by it and read both ends. Which city has the smallest gap, and does its `a_tracts` and `d_tracts` count make that number worth anything?
- `tracts.csv` has `grade` beside `total` and `pct_black`. Group by `city` and `grade` and compare the population under each letter. Which cities graded most of their people C, and does the amount of D-graded ground track the size of the gap?
- `fig_mix.csv` carries `own_share`, how much of a tract's graded land actually carries the letter the tract was given. Sort it and find the tracts where it is lowest. Those are the rows any map of this data draws wrong.
- `fig_quote.csv` holds one appraiser's own words about one neighborhood, with the `label` and `dated` of the sheet. Read it, then look up what that neighborhood is like now.
- **Stretch.** *Mapping Inequality* publishes the area description sheets for every graded city. Pick a city in `cities.csv` and read three D sheets and three A sheets, and count how many name a group of people rather than a condition of the buildings.
- **Stretch.** The 2020 census also publishes Hispanic origin and household income by tract. Attach one of them to `tracts.csv` by `GE0ID` and ask whether the within-city gap survives the change of variable.

07 Sources

Data

- Nelson, R. K., Winling, L., Marciano, R., Connolly, N. D. B., et al. *Mapping Inequality: Redlining in New Deal America*. Digital Scholarship Lab, University of Richmond, CC BY-NC-SA. <https://dsl.richmond.edu/panorama/redlining/> — **the HOLC polygons, the grade colors and the appraisal text drawn and quoted above are theirs**. Retrieved 2026-08-10: the national area file (`static/mappinginequality.json`, 10,563,682 bytes, 10,154 areas across 314 city surveys, of which 9,324 carry a grade A to D) and the area description sheet for OHCleveland1939 area D9. The maps and forms underneath are 1930s US federal government records.
- U.S. Census Bureau, 2020 Decennial Census, the P.L. 94-171 redistricting file, one state at a time. https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/
- U.S. Census Bureau, TIGER/Line 2020 census tract boundaries — the outlines drawn over the 1939 areas in Figure 2.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

```
cities.csv,fig_holc_attr.csv,fig_holc_meta.csv,fig_holc_rings.csv,
fig_mix.csv,fig_quote.csv,fig_source.csv,fig_zoom_attr.csv,fig_zoom_holc.csv,
fig_zoom_tracts.csv,fig_zoom_win.csv,tracts.csv
```

Academic literature

- Aaronson, D., Hartley, D. and Mazumder, B. (2021). "The Effects of the 1930s HOLC 'Redlining' Maps." *American Economic Journal: Economic Policy* 13(4): 355-392.
- Rothstein, R. (2017). *The Color of Law: A Forgotten History of How Our Government Segregated America*. Liveright.

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original sources, without any starting files. Please do the following and show your work.

THE FIRST SOURCE. "Mapping Inequality: Redlining in New Deal America", Digital Scholarship Lab, University of Richmond (CC BY-NC-SA). One GeoJSON file, about 10.5 MB, holding every area drawn on the Home Owners' Loan Corporation's 1930s security maps:

<https://dsl.richmond.edu/panorama/redlining/static/mappinginequality.json>

No account or key is needed. Each feature is one appraisal area with its city, state, and grade – 10,154 areas, of which 9,324 carry a grade A through D. The grade field arrives padded with spaces: the same grade appears as "A", "A ", and " A". Trim it before filtering, because keeping only A through D without trimming looks like a clean subset and silently drops areas.

THE SECOND SOURCE. U.S. Census Bureau, TIGER/Line 2020 census-tract boundaries, one zip per state, at

https://www2.census.gov/geo/tiger/TIGER2020/TRACT/tl_2020_<FIPS>_tract.zip

for six states: California (06), Illinois (17), Michigan (26), Missouri (29), Ohio (39), Pennsylvania (42). Every tract polygon carries an internal point the Bureau guarantees to lie inside the tract.

THE THIRD SOURCE. 2020 Census P.L. 94-171 redistricting data for the same six states – table P1 has each tract's total, white, and Black population. One zip per state, named like Michigan's:

https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File-PL_94-171/Michigan/mi2020.pl.zip

The files inside are pipe-delimited segments that join on a record number. Read that number as text on both sides – it is zero-padded, and a numeric read makes the join silently match nothing.

WHAT TO BUILD.

1. A grade for each 2020 tract: the tract takes the grade of the graded area its internal point falls inside, first match wins. Keep the tracts that get one.
2. Those graded tracts joined to their 2020 population counts, with percent Black per tract.
3. For every city holding both an A tract and a D tract, the percent Black of the pooled population on each side, and the gap between the two.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- 10,154 areas in the map file, 9,324 of them graded A through D
- 4,489 tracts carrying a grade across the six states
- 33 cities with both an A and a D tract; the widest gap is Cleveland, whose once-A tracts hold 79,609 people and whose once-D tracts hold 100,969

Mapping Inequality keeps adding newly digitized cities; a larger area count means the project grew, not that you made an error.